

Tracking structural cues or relying on probabilistic inference in Turkish agreement?

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Today's talk

How is **structural information** used to encode and access linguistic representations in memory?

Using **agreement attraction** in Turkish as a test case;

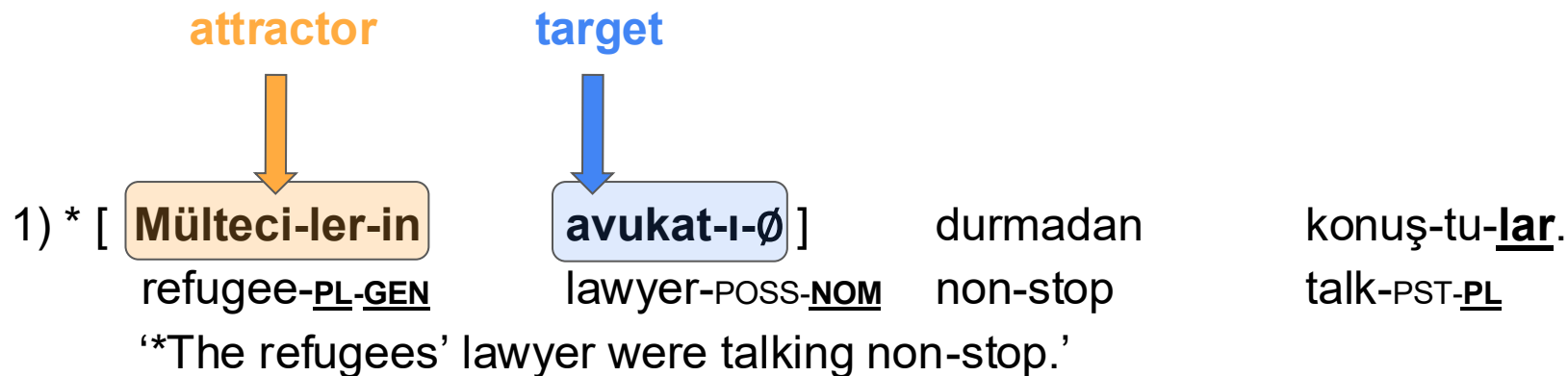


We show evidence for the role of **syntactic positional associations**.



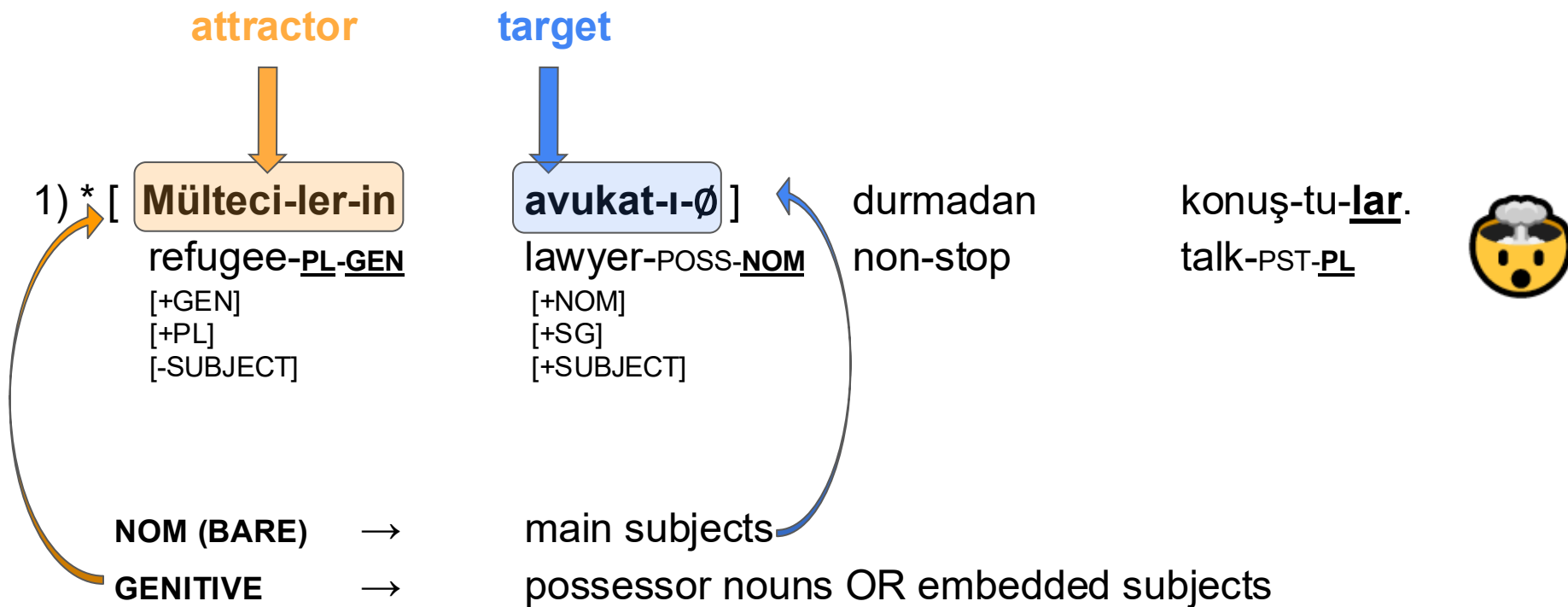
Abstract syntactic cues shape how linguistic representations are **encoded** and **accessed in memory**.

Agreement attraction in Turkish

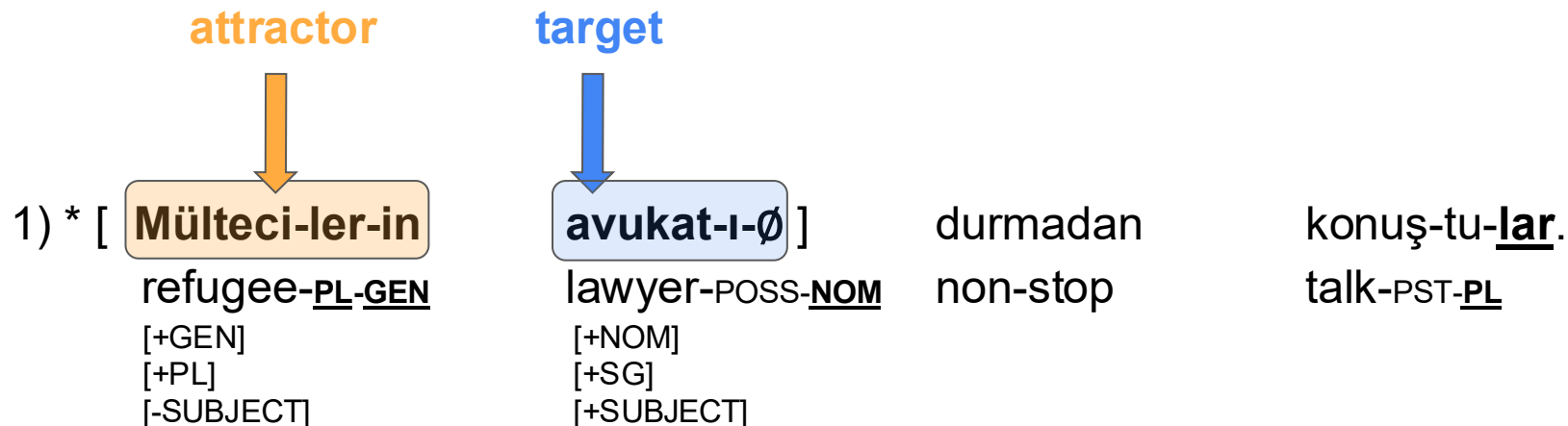


Ungrammatical sentences with a **plural**, **GENITIVE-marked attractor** occasionally accepted as grammatical

Agreement attraction in Turkish



Agreement attraction in Turkish



TODAY: What type of structural information modulates memory interference?

①

Surface structural similarity to the subject (CASE)



②

Syntactic associations with SUBJECT PHRASE

Experiment 1: Overview

Surface similarity to the target *or* **syntactic associations** with the subject?

A speeded acceptability judgment task;

2x3 within-subject design

N_{participant} = 92 Turkish speakers

N_{item} = 24

N_{filler} = 48 (16 ungrammatical)

Experiment 1:

Surface similarity or syntactic associations?

ATTRACTOR WITHIN SUBJECT

Dün / Yesterday...

TARGET MATCH

[mülteci-nin avukat-lar-ı]
 refugee-GEN lawyer-PL-POSS

... duruşma-ya ...
 trial-DAT

katıldı-lar.
 joined-PL

ATTRACTOR MATCH

* [mülteci-ler-in avukat-ı]
 refugee-PL-GEN lawyer-POSS

...

katıldı-lar.
 joined-PL

NO MATCH

* [mülteci-nin avukat-ı]
 refugee-GEN lawyer-POSS

...

katıldı-lar.
 joined-PL

ATTRACTOR WITHIN OBJECT

TARGET MATCH

avukat-lar
 lawyer-PL

... [mülteci-nin duruşma-sın-a]
 refugee-GEN trial-POSS-DAT

...

katıldı-lar.
 joined-PL

ATTRACTOR MATCH

* avukat
 lawyer

* [mülteci-ler-in
 refugee-PL-GEN

...

katıldı-lar.
 joined-PL

NO MATCH

* avukat
 lawyer

[mülteci-nin
 refugee-GEN

...

katıldı-lar.
 joined-PL

'Yesterday, the (refugee's) lawyers joined the (refugee's) trial.'

Experiment 1:

Surface similarity or syntactic associations?

ATTRACTOR WITHIN SUBJECT

Dün / Yesterday...

TARGET MATCH	[mülteci- <u>nin</u> refugee- <u>GEN</u>]	[avukat- <u>lar-ı</u> lawyer- <u>PL-POSS</u>]
ATTRACTOR MATCH	* [mülteci- <u>ler-in</u> refugee- <u>PL-GEN</u>]	[avukat-ı lawyer-POSS]
NO MATCH	* [mülteci- <u>nin</u> refugee- <u>GEN</u>]	[avukat-ı lawyer-POSS]

ATTRACTOR WITHIN OBJECT

TARGET MATCH	[avukat- <u>lar</u> lawyer- <u>PL</u>]	... [mülteci- <u>nin</u> ...]
ATTRACTOR MATCH	* avukat lawyer	[mülteci- <u>ler-in</u> refugee- <u>PL-GEN</u>]
NO MATCH	* avukat lawyer	[mülteci- <u>nin</u> refugee- <u>GEN</u>]

PREDICTIONS

Surface similarity;

both attractors look like SUBJECT

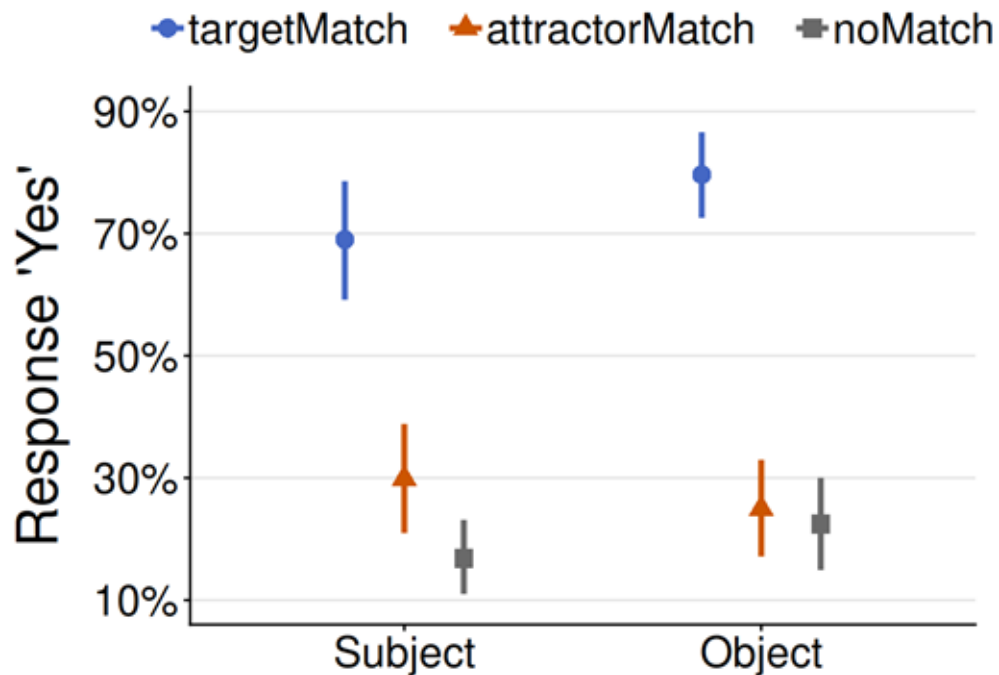
SUBJECT $\approx$ OBJECT

Syntactic associations;

attractor within SUBJECT is syntactically more similar to SUBJECT

SUBJECT $>$ OBJECT

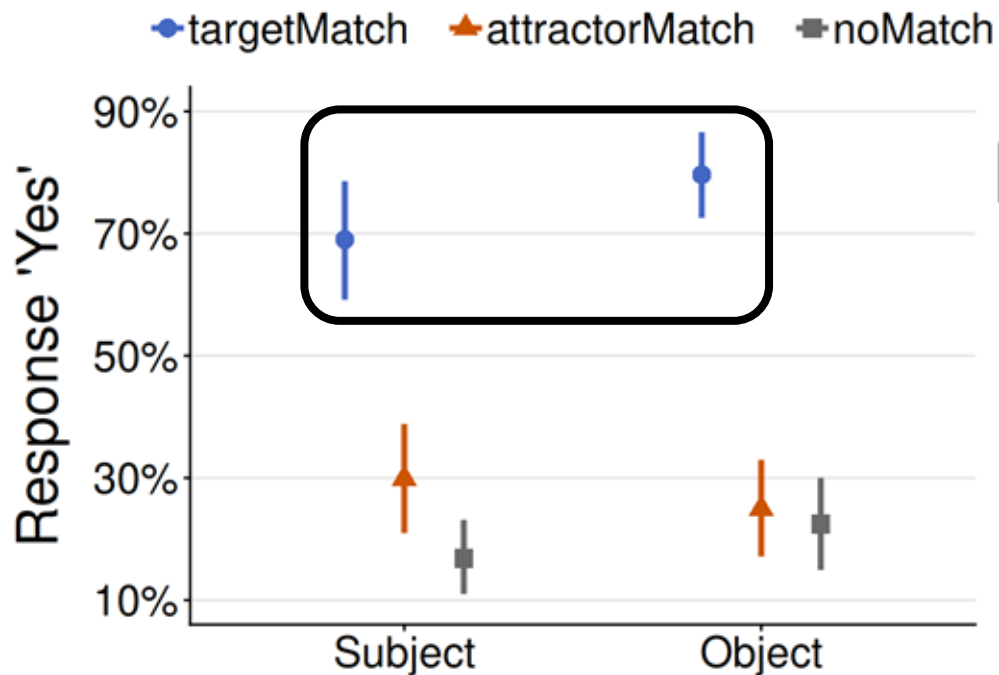
Experiment 1: Results



Error bars show 89% HPD

$N_{\text{participant}} = 92$

Experiment 1: Results



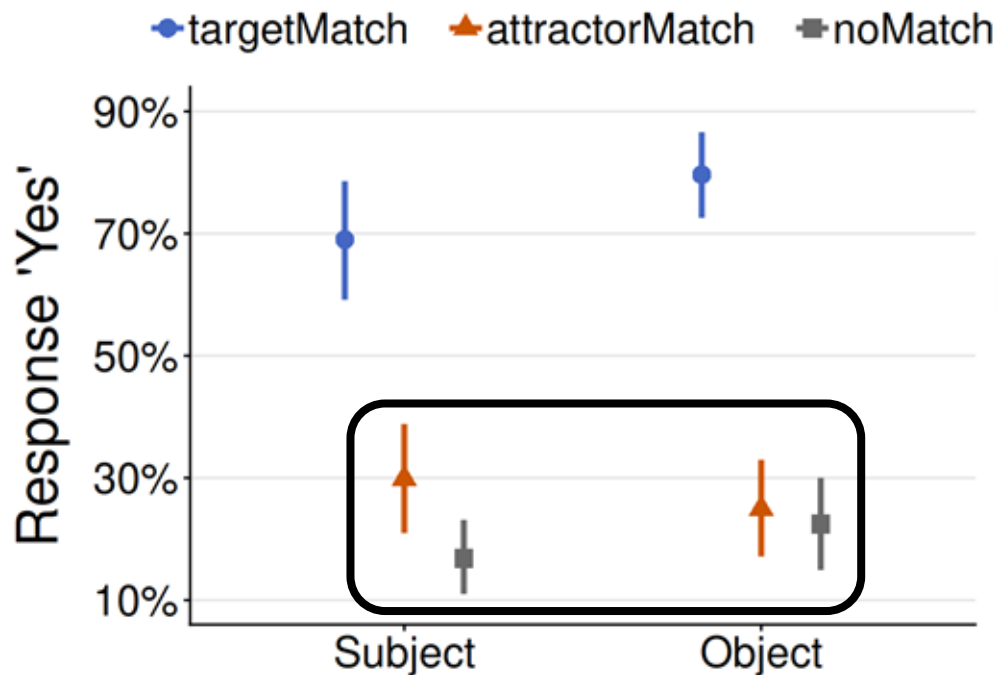
Grammaticality

$\beta = .77$ [.61, .93], $P(\beta > 0) > 0.99$

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Experiment 1: Results



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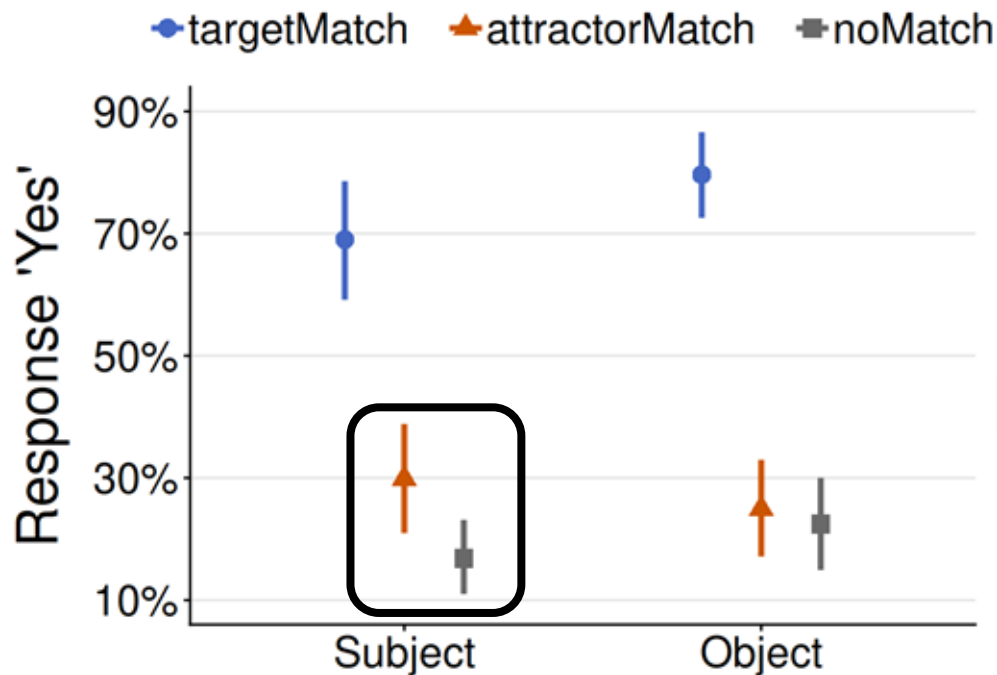
Grammaticality

$\beta = .77$ [.61, .93], $P(\beta > 0) > 0.99$



Position*Attraction $\beta = .16$ [.02, .29], $P(\beta > 0) = .97$

Experiment 1: Results



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Grammaticality

$\beta = .77$ [.61, .93], $P(\beta > 0) > 0.99$

Position*Attraction

$\beta = .16$ [.02, .29], $P(\beta > 0) = .97$



Subject: Attraction

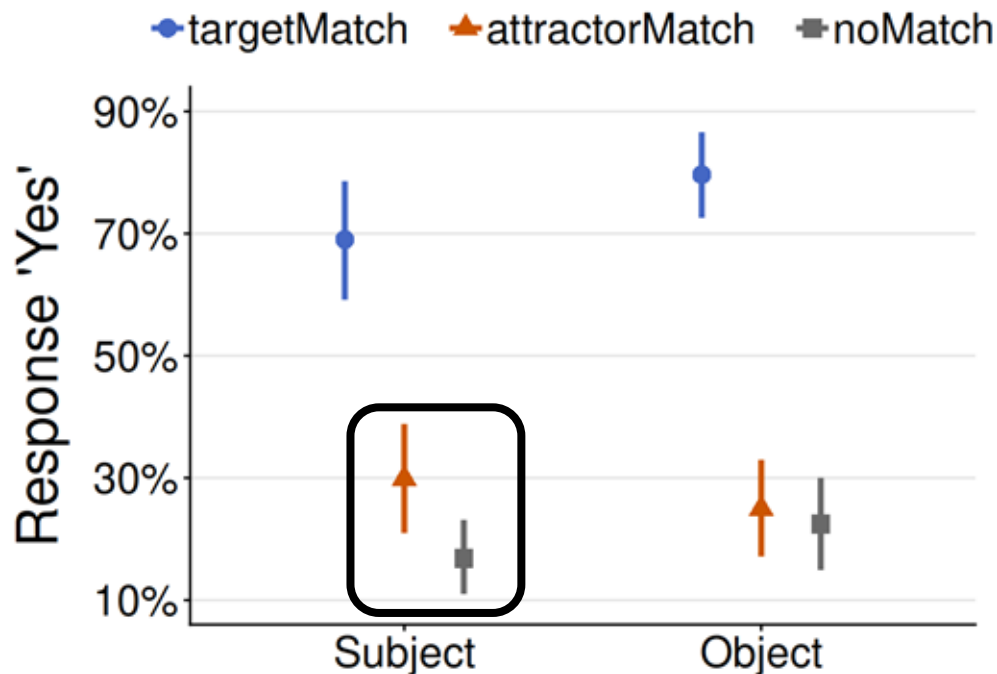
$\beta = .37$ [.19, .55], $P(\beta > 0) = .99$



Object: Attraction

$\beta = .07$ [-.12, .25], $P(\beta > 0) = .73$

Experiment 1: Results



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Object: Attraction

$\beta = .07$ [-.12, .25], $P(\beta > 0) = .73$



**Syntactic association with the subject
leads to attraction**

Experiment 2: Overview

Does attraction extend to other **attractors with similar syntactic associations**?

A speeded acceptability judgment task;

2x3 within-subject design

$N_{\text{participant}}$ = 58 Turkish speakers

N_{item} = 24

N_{filler} = 48 (16 ungrammatical)



BARE attractors

Experiment 2: Bare attractors?

GENITIVE ATTRACTOR = ATTRACTOR WITHIN SUBJECT IN E1

TARGET	Mülteci- nin	avukat- lar-ı	duruşmada	durmadan
MATCH	refugee- <u>GEN</u>	lawyer- <u>PL</u> -POSS	at.trial	non-stop
ATTRACTOR	* Mülteci- ler-in	avukat-ı		
MATCH	refugee- <u>PL-GEN</u>	lawyer-POSS		
NO MATCH	* Mülteci- nin	avukat-ı		
	refugee- <u>GEN</u>	lawyer-POSS		

konuştu-**lar**.
talked-PL

konuştu-**lar**
talked-PL

konuştu-**lar**.
talked-PL

BARE ATTRACTOR

TARGET	Mülteci	avukat- lar-ı
MATCH	refugee	lawyer- <u>PL</u> -POSS
ATTRACTOR	* Mülteci- ler	avukat-ı
MATCH	refugee- <u>PL</u>	lawyer-POSS
NO MATCH	* Mülteci	avukat-ı
	refugee	lawyer-POSS

konuştu-**lar**.
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TARGET	Mülteci- <u>nin</u>	avukat- <u>lar</u> -ı
MATCH	refugee- <u>GEN</u>	lawyer- <u>PL</u> -POSS
ATTRACTOR	* Mülteci- <u>ler</u> - <u>in</u>	avukat-ı
MATCH	refugee- <u>PL</u> - <u>GEN</u>	lawyer-POSS
NO MATCH	* Mülteci- <u>nin</u>	avukat-ı
	refugee- <u>GEN</u>	lawyer-POSS

BARE ATTRACTOR

TARGET	Mülteci	avukat- <u>lar</u> -ı
MATCH	refugee	lawyer- <u>PL</u> -POSS
ATTRACTOR	* Mülteci- <u>ler</u>	avukat-ı
MATCH	refugee- <u>PL</u>	lawyer-POSS
NO MATCH	* Mülteci	avukat-ı
	refugee	lawyer-POSS

*Probabilistic inference for
alternatives with high frequency*

likely! $P(\text{target.PL} \mid \text{attractor.PL})$

*Plural target assumed in ATTRACTOR MATCH
→ attraction*

Experiment 2: Bare attractors?

GENITIVE ATTRACTOR = ATTRACTOR WITHIN SUBJECT IN E1

TARGET	Mülteci- <u>nin</u>	avukat- <u>lar-ı</u>
MATCH	refugee- <u>GEN</u>	lawyer- <u>PL</u> -POSS
ATTRACTOR	* Mülteci- <u>ler-in</u>	avukat-ı
MATCH	refugee- <u>PL-GEN</u>	lawyer-POSS
NO MATCH	* Mülteci- <u>nin</u>	avukat-ı
	refugee- <u>GEN</u>	lawyer-POSS

$$P(\text{target.PL} \mid \text{attractor.PL}) > P(\text{target.PL} \mid \text{attractor.SG})$$



BARE ATTRACTOR

TARGET	Mülteci	avukat- <u>lar-ı</u>
MATCH	refugee	lawyer- <u>PL</u> -POSS
ATTRACTOR	* Mülteci- <u>ler</u>	avukat-ı
MATCH	refugee- <u>PL</u>	lawyer-POSS
NO MATCH	* Mülteci	avukat-ı
	refugee	lawyer-POSS

ungrammatical?!

$$[\text{attractor.PL} + \text{target.PL}]$$



Experiment 2: Bare attractors?

GENITIVE ATTRACTOR = ATTRACTOR WITHIN SUBJECT IN E1

TARGET	Mülteci- <u>nin</u>	avukat- <u>lar</u> -ı
MATCH	refugee- <u>GEN</u>	lawyer- <u>PL</u> -POSS
ATTRACTOR	* Mülteci- <u>ler-in</u>	avukat-ı
MATCH	refugee- <u>PL-GEN</u>	lawyer-POSS
NO MATCH	* Mülteci- <u>nin</u>	avukat-ı
	refugee- <u>GEN</u>	lawyer-POSS

BARE ATTRACTOR

TARGET	Mülteci	avukat- <u>lar</u> -ı
MATCH	refugee	lawyer- <u>PL</u> -POSS
ATTRACTOR	* Mülteci- <u>ler</u>	avukat-ı
MATCH	refugee- <u>PL</u>	lawyer-POSS
NO MATCH	* Mülteci	avukat-ı
	refugee	lawyer-POSS

PREDICTIONS

Syntactic associations;

both attractors are comparable and syntactically similar to SUBJECT

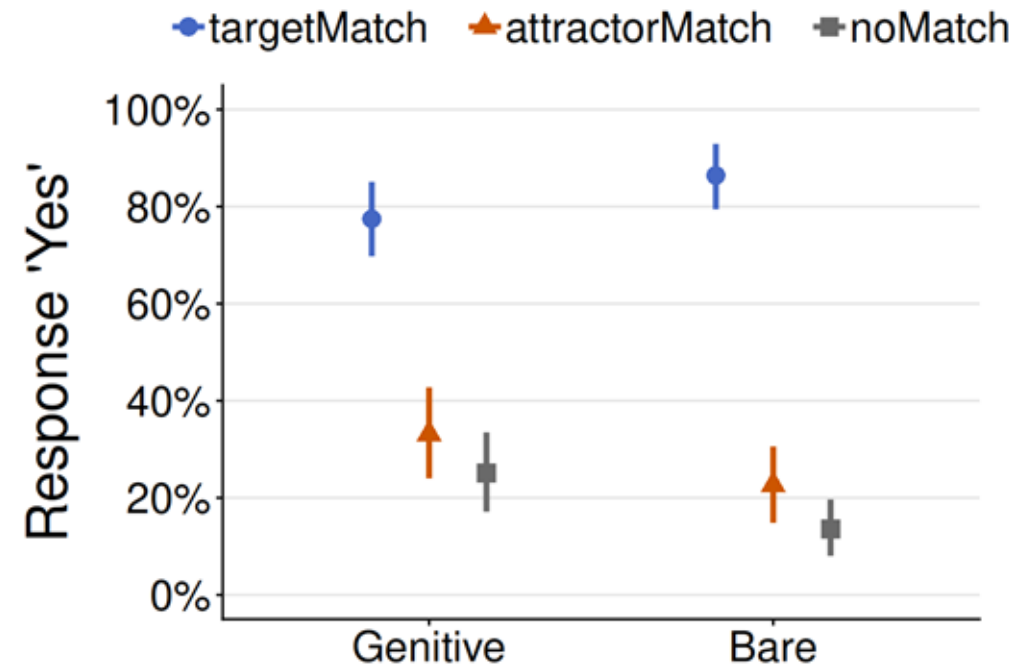
GEN $\approx$ BARE

Probabilistic inference

target in GEN is likely to be corrected to
a plural form

GEN $>$ BARE

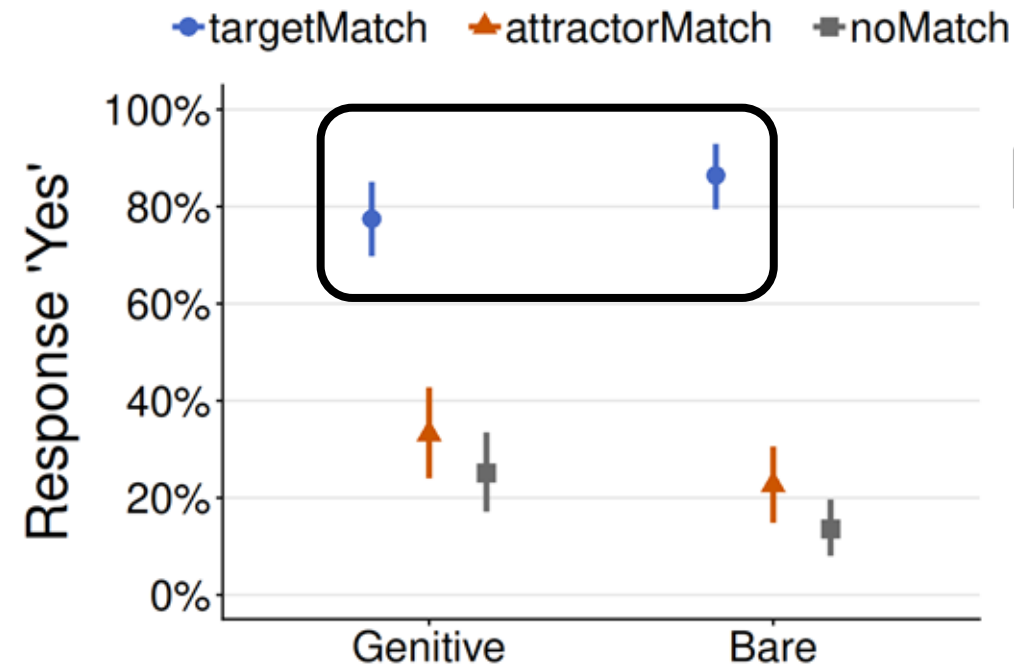
Experiment 2: Results



Error bars show 89% HPD

$N_{\text{participant}} = 58$

Experiment 2: Results



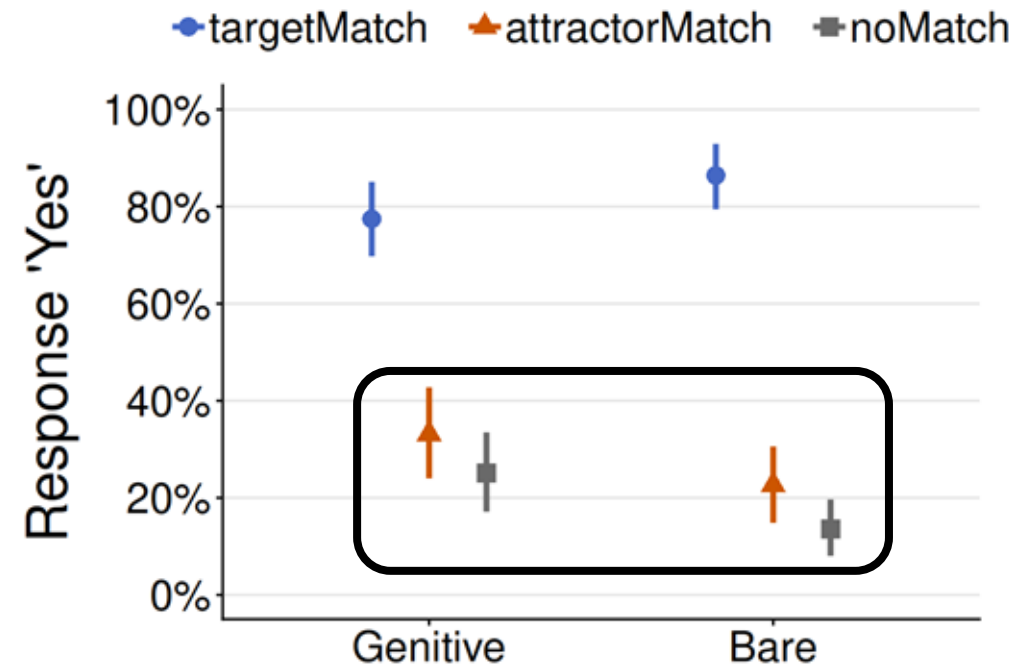
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Attraction

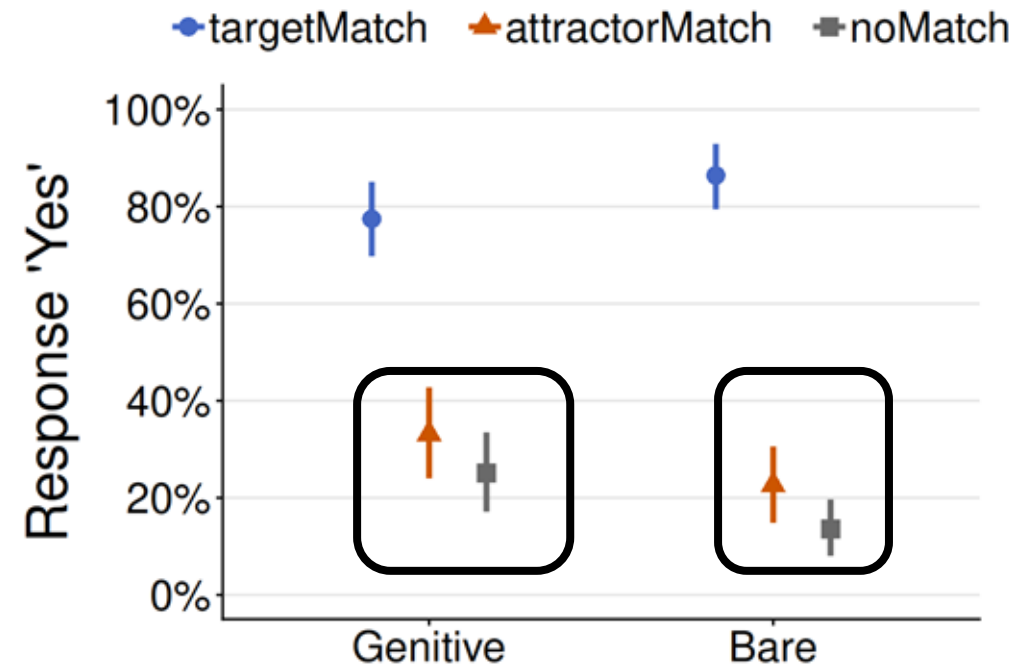
$\beta = .26$ [.09, .42], $P(\beta > 0) = .99$



Case * Attraction

$\beta = -.05$ [-.23, .14], $P(\beta > 0) = .34$

Experiment 2: Results



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Genitive: Attraction

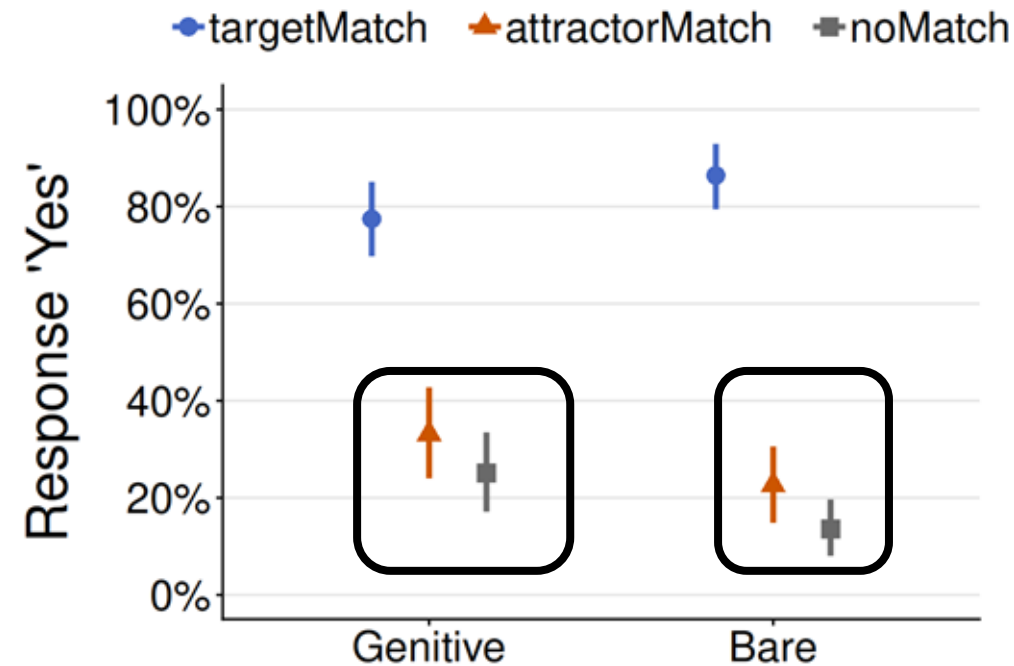
$\beta = .20$ [-.03, .42], $P(\beta > 0) = .92$



Bare: Attraction

$\beta = .31$ [.06, .56], $P(\beta > 0) = .98$

Experiment 2: Results



Error bars show 89% HPD

$N_{\text{participant}} = 58$

Grammaticality

$\beta = .92$ [.76, 1.09], $P(\beta > 0) > 1$

Attraction

$\beta = .26$ [.09, .42], $P(\beta > 0) = .99$

Case * Attraction

$\beta = -.05$ [-.23, .14], $P(\beta > 0) = .34$

Genitive: Attraction

$\beta = .20$ [-.03, .42], $P(\beta > 0) = .92$

Bare: Attraction

$\beta = .31$ [.06, .56], $P(\beta > 0) = .98$



Attraction extends to bare attractors

Summary

In two experiments on **agreement attraction in Turkish**, we

- ✓ replicated attraction with **GENITIVE attractors embedded within the SUBJECT**
- ✓ presented evidence for attraction with **BARE attractors**
- ✗ found **no evidence** for attraction with **GENITIVE attractors within the OBJECT**

Main take-away:

Memory access in Turkish gated by **syntactic associations with the SUBJECT**

→ role of surface cues such as case & probabilistic inference?

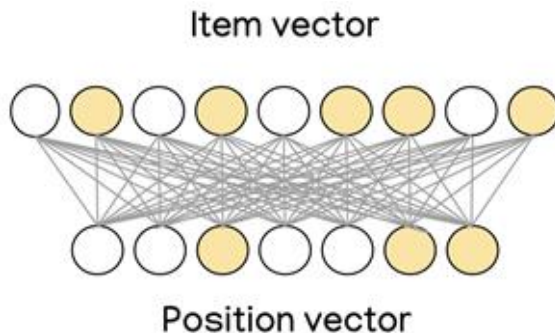
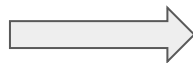
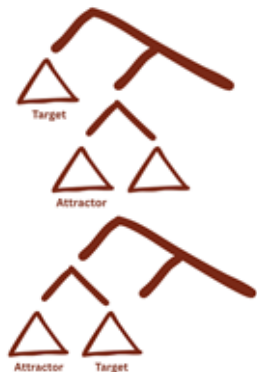
Potential role for surface cues? → attractors with non-subject cases

Discussion



How to model associations between lexical items and syntactic positions in memory?

Items maintained through transient bindings with their syntactic positions,
with memory access guided by cues linked to these positions



(Keshev et al., 2025; Bakay et al., 2026)

Attractors that are syntactically associated with the subject → **memory interference**

Discussion



How are different sources of information used **crosslinguistically**?

Syntactic associations, surface cues, probabilistic inference, semantic information...

Structure in LLMs: Attention allocation in transformer-based BERT in Turkish;

More attention to PLURAL attractor; increased with overt **GENITIVE** within the **SUBJECT**

(Ryu & Lewis, 2025)

THANK YOU!

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